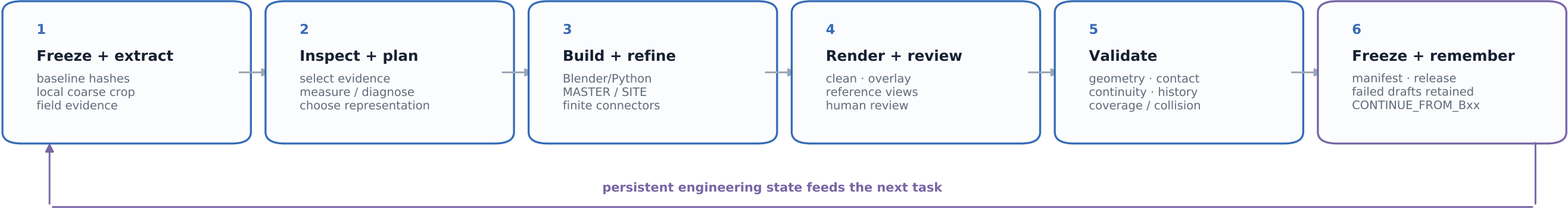


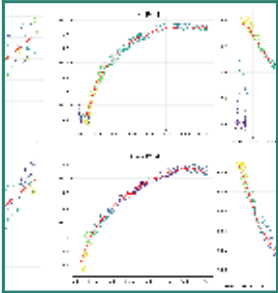
Canonical reconstruction loop with task-specific engineering operators

The historical batches share a stable core loop; profile fitting, routing, coverage, collision, and human-review operators are enabled only when the task requires them.



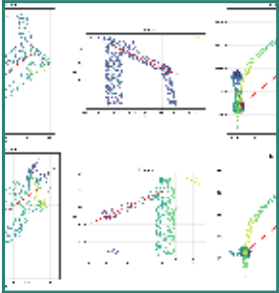
Task-specific operators — invoked as needed, not mandatory stages

Profile / measurement fitting



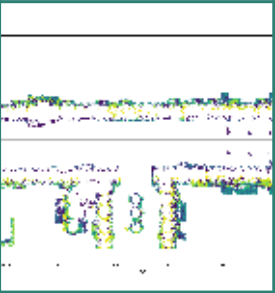
B23: point-cloud profiles guide curved lead and shared component line

Representation preflight



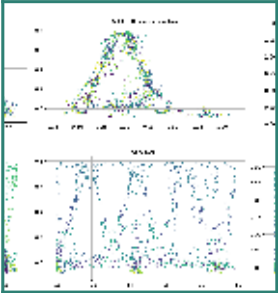
B23: observed line hypothesis is contradicted by the observed data

Route / topology diagnosis



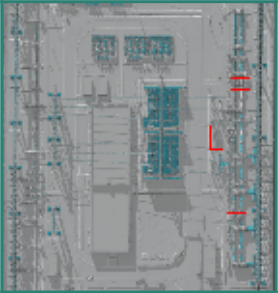
B20: bus route discontinuity and turn geometry before construction

Section / structure fitting



B25: cross-sections and repeated brace spans

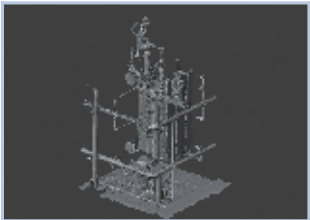
Human spatial feedback



D38: stored review markup becomes a coordinate-grounded correction task

Every batch leaves reviewable result evidence

Reference



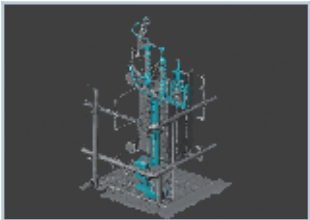
registered source/coarse evidence in the comparison view

Clean



component model without the comparison overlay

Overlay



same-view model/reference comparison

Process evidence is part of the result

plans · profiles · preflight diagnostics · component libraries
clean/overlay/reference renders · validation ledgers · manifests
failed revisions · continuation notes

Figure 11 · The common core is supported by repeated prepare/extract/build/validate/release stages; specialized operators vary by equipment and failure mode. Intermediate artifacts externalize geometric hypotheses and revisions rather than serving only as presentation graphics.